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KRISTU JAYANTI INSTITUTE OF TECHNOLOGY

PROJECT REPORT ON

"TIMORA – KERALA'S SMART TIMBER MARKETPLACE"

Submitted in partial fulfillment of the requirements for Enterprise Computing Project Lab for 3rd Semester

Master of Computer Applications (MCA)

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CERTIFICATE

This is to certify that the project titled ***"TIMORA – Kerala's Smart Timber Marketplace"*** has been satisfactorily completed by **ROHAN K CHERIAN [25MCAD34]** in partial fulfillment of the requirements for ***Enterprise Computing Project*** with course code **25MCC7P231**, for the 3rd Semester MCA course during the academic semester from July 2026 to November 2026 as prescribed by Kristu Jayanti University.

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ABSTRACT

Kerala's timber trade is economically vital but fragmented. Opaque brokerage markups of 15% to 30%, arbitrary volume measurements, complicated Form IV transit-permit compliance and the absence of a central discovery platform hurt growers, sawmills, carpenters, transporters and buyers alike. This project presents **TIMORA**, a smart timber marketplace for the 14 districts of Kerala that connects these participants directly.

TIMORA is built as a Turborepo monorepo with a React 18 and Vite progressive web application styled with Tailwind CSS, an Express.js REST API written in TypeScript, a Supabase-hosted PostgreSQL database, an Expo mobile shell and a shared package containing the Kerala Quarter-Girth Hoppus and rectangular CFT formulas and Zod validation schemas. The platform offers a raw timber stock exchange for six native species, a dual-mode CFT calculator, verified directories of sawmills, carpenters and transporters, six role-isolated dashboards including a Super Admin moderation cell, Form IV verification badges, support ticketing with real-time bilingual notifications, and English and Malayalam interfaces.

Security is provided by Argon2id PIN hashing, seven-day JWT sessions with a 25-minute inactivity timeout, Helmet headers, parameterised queries and rate limiting. The system is deployed on Vercel and Supabase and verified by 186 passing automated Vitest tests. TIMORA brings transparency, standard measurement and regulatory awareness to Kerala's timber trade.

Keywords: Timber marketplace, Hoppus rule, CFT, React, Express, PostgreSQL, Supabase, Vercel, Progressive Web App, Kerala.

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CHAPTER NO 1 INTRODUCTION

1.1 Problem Statement

Kerala's domestic timber trade is a foundational economic driver in southern India. It supplies residential roof construction, heritage temple woodwork, architectural joinery and estate agriculture. Despite this importance, the market remains severely fragmented. Growers, sawmill owners, master carpenters and haulage operators work in regional silos with no common platform for pricing, discovery or trust.

Four recurring problems define the legacy market:

- **Brokerage exploitation:** intermediaries typically levy unrecorded markups of 15% to 30% between tree growers and end buyers, without offering quality assurance or legal indemnity.
- **Measurement disputes:** quarter-girth formulas are applied arbitrarily, without a standardised Hoppus conversion. This leads to repeated commercial friction over cubic-feet (CFT) calculations and taper deductions.
- **Legal compliance hurdles:** moving timber across Kerala's checkposts requires Form IV / Form III transit permits issued under the Kerala Forest Produce Transit Rules, 1975. Non-compliance results in vehicle seizures and long delays.
- **Lack of direct discovery:** smallholder growers, licensed sawmills, carpenters and heavy-vehicle operators have no centralised, transparent view of available stock, rates and services.

The problem addressed by this project is therefore: **how can a single trusted digital platform connect all participants of Kerala's timber trade, standardise volume measurement, surface transit-permit compliance, and remove opaque intermediaries?**

1.2 Project Description

TIMORA™ (tagline: "Timber · People · Possibilities") is a web-based smart timber marketplace for the 14 districts of Kerala. It provides a unified digital marketplace and logistics directory connecting timber buyers, individual sellers, sawmill owners, master carpenters and transport operators. The platform enforces pure raw-stock catalogue validation, digital Form IV compliance badges, automated Kerala Hoppus-rule volume calculation and role-isolated dashboards, with moderation by a Super Admin cell.

The system is delivered as a Turborepo monorepo containing a React single-page progressive web application, an Express.js REST API backed by Supabase-hosted PostgreSQL, an Expo mobile application shell, and a shared TypeScript package. The production system is live on Vercel and is verified by an automated suite of 186 Vitest tests.

Objectives

- Provide a transparent marketplace where growers and sellers list raw timber lots directly to buyers.
- Standardise volume measurement through a built-in Kerala Hoppus-rule and rectangular CFT calculator shared across all clients and the server.
- Display Forest Form IV transit-pass verification so that buyers can identify compliant lots.
- Give every market role (buyer, seller, mill owner, carpenter, transport owner, admin) a dedicated, access-controlled dashboard.
- Offer a moderated support-ticket system with real-time in-dashboard notifications.
- Deliver a responsive, installable, bilingual (English and Malayalam) experience suitable for low-end devices and unreliable networks.

Module Description

The platform is organised into the functional modules summarised in Table 1.1 and described below.

Table 1.1: Summary of Platform Modules

No.	Module	Route	Purpose
1	Raw Timber Stock Exchange	/marketplace	Discover and filter raw timber lots
2	Hoppus CFT Calculator	/calculator	Compute volume and cost
3	Licensed Sawmills Directory	/mills	Find verified sawmills
4	Master Wood Artisans Directory	/carpenters	Find verified carpenters
5	Timber Logistics & Haulage	/transport	Find compliant transport operators
6	Multi-Role Dashboards	/dashboard/*, /admin	Role-specific operations and moderation
7	Support Ticketing & Notifications	Dashboard	Dispute handling and alerts
8	Authentication & Security	/login	Phone + PIN login, JWT sessions

Module 1: Raw Timber Stock Exchange

The core marketplace enables discovery of raw timber lots across Kerala's 14 districts. It supports six dominant native species: Nilambur Teak , African Mahogany, Old-Growth Anjili / Wild Jack, Estate Rubberwood, Heritage Veetti / Rosewood and Varikka Plavu / Jackfruit . Capabilities include species pill filters, district sorting, quantity sorting (CFT / tonnes), Forest Form IV verification badges, and an automated spam detector that rejects non-timber inquiries such as sofas, plywood or processed tables.

Module 2: Kerala Standard Hoppus Rule CFT Calculator

An interactive dual-mode volumetric calculator conforming to the Kerala Forest and PWD conventions. Round logs are computed with the Quarter-Girth Hoppus formula, and sawn beams and planks with the rectangular CFT formula. The module also converts CFT to cubic metres and estimates cost from prevailing market rates.

Module 3: Licensed Sawmills Directory

A verified directory of licensed horizontal and vertical band-saw mills across Ernakulam, Kottayam, Pathanamthitta, Thrissur, Kozhikode and Palakkad. It shows sawing rates per CFT (₹65 to ₹85), daily capacity (up to 400 CFT per day), kiln-seasoning availability, WhatsApp click-to-chat and verified customer reviews.

Module 4: Master Wood Artisans Directory

Curated profiles of verified Kerala master carpenters specialising in traditional Nalukettu joinery, temple timber work, heavy roof trusses, solid teak entrance doors and wood-seasoning inspection.

Module 5: Timber Logistics and Haulage

A directory of heavy timber transit operators with self-loading hydraulic cranes, multi-axle trailers and Form IV compliant carriers. Pre-inspected fleet verification helps prevent transit delays.

Module 6: Multi-Role Portals and Isolated Dashboards

- **Buyer Dashboard (/dashboard/buyer):** review live orders, track confirmed consignments, submit verified ratings and communicate with sellers.
- **Depot Seller Dashboard (/dashboard/seller):** manage inventory, upload multi-angle photographs, adjust CFT prices and review enquiries.
- **Timber Mill Owner Dashboard (/dashboard/mill-owner):** post sawn timber batches (planks, squared beams, battens), record daily sawing throughput and accept sawing orders.
- **Carpenter Dashboard (/dashboard/carpenter):** receive joinery commissions, showcase specialisations and manage ongoing residential projects.
- **Transport Owner Dashboard (/dashboard/transport):** register the haulage fleet (tonnage, hydraulic crane) and review load-booking enquiries.
- **Super Admin Control Cell (/admin):** moderate pending business applications, verify sawmill licences, review audit-log entries and resolve support grievances.

Module 7: Support Grievance Ticketing and Live Notifications

A centralised ticketing cell lets users raise disputes about wood quality, measurement discrepancies or checkpost delays. Each ticket receives a sequential reference number such as TKT-2026-0042. When an administrator resolves a ticket, a resolution notification is pushed into the user's dashboard feed.

Module 8: Authentication, Security and Loading Experience

Users authenticate with a mobile number and a 6-digit PIN and receive a JWT bearer token valid for seven days. A 25-minute inactivity timeout triggers a PIN re-authentication modal to protect shared depot computers. Every page uses loading skeletons, and action buttons disable and show a spinner with a status label ("Signing in…", "Publishing…", "Saving…") to prevent double submissions.

Objectives

The primary objective of TIMORA is to develop a smart, transparent and user-friendly digital marketplace that connects timber buyers, individual sellers, sawmill owners, carpenters and transport operators across the 14 districts of Kerala. The project aims to reduce the dependency on informal brokers and establish a direct connection between timber growers and buyers, ensuring better price transparency and accessibility. It also aims to standardise timber volume calculations through an integrated Kerala Quarter-Girth Hoppus and rectangular Cubic Feet (CFT) calculator, minimising measurement errors and disputes between buyers and sellers. Another important objective is to improve transparency in timber transportation by providing Forest Form IV transit-permit verification details and verified business profiles. The system also aims to provide separate role-based dashboards for different users, enabling them to manage their listings, orders, services and business activities efficiently. In addition, TIMORA focuses on providing a secure authentication mechanism, administrator-based verification, customer support ticketing and real-time notifications. The platform is designed to offer a responsive and installable Progressive Web Application with English and Malayalam language support, making timber-related services more accessible to users throughout Kerala.

Scope

The scope of TIMORA covers the development of a centralised digital timber marketplace designed specifically for the timber industry in Kerala. The platform enables users across all 14 districts to discover, list and manage raw timber products through an online marketplace with filtering options based on timber species, district, quantity and price. It includes an integrated Hoppus and rectangular CFT calculator to calculate timber volume and estimate its cost using standard measurement formulas. The system also provides dedicated directories for licensed sawmills, master carpenters and timber transportation operators, allowing users to identify and connect with relevant service providers. Different user roles, including buyers, sellers, mill owners, carpenters, transport owners and administrators, are provided with separate dashboards and access permissions to manage their respective activities. The scope further includes business verification, timber listing moderation, Form IV compliance indicators, order management, support ticket handling and in-dashboard notifications. Security features such as PIN-based authentication, JWT sessions, input validation and role-based access control are incorporated to protect user information and system operations. The application supports English and Malayalam interfaces and is designed to function across desktop computers, tablets and mobile devices, including low-end devices with intermittent network connectivity. The current implementation focuses on Kerala's domestic timber marketplace, providing a foundation for future enhancements such as online payments, shipment tracking, automated permit verification, market price analytics and expansion into neighbouring states.

CHAPTER NO 2 SYSTEM STUDY

2.1 Existing System

At present, Kerala's timber trade runs largely on personal contacts, telephone calls and informal brokers. Growers learn about buyers through intermediaries, who set the price, charge an undeclared margin and control the flow of information. Sawmills and carpenters are found by word of mouth, and transporters are arranged case by case.

The limitations of the existing system are listed below.

- No price transparency; the grower rarely knows the final market value of a lot.
- Brokerage margins of 15% to 30% that are neither recorded nor justified by quality assurance.
- No standard volume calculation, so the buyer and seller frequently dispute CFT and taper deductions.
- Transit-permit status (Form IV / Form III) is verified manually, often only at checkposts, leading to seizures and delays.
- No verified directory of sawmills, carpenters or haulage operators, and no review or rating mechanism.
- No formal grievance process for quality or measurement disputes.
- Generic classified portals do not understand species, CFT or transit rules and accept irrelevant listings.

2.2 Proposed System

The proposed system, TIMORA, replaces informal brokerage with a moderated, role-based digital marketplace. Sellers publish lots directly, an administrator verifies businesses and listings, and buyers browse transparent, filterable stock with standardised CFT figures and compliance badges.

The main features of the proposed system are:

- Direct grower-to-buyer discovery with species, district and quantity filters.
- Embedded Hoppus and rectangular CFT calculators from a single shared library, so web, mobile and server compute identical results.
- Form IV transit-pass verification badges and sawmill licence verification by moderators.
- Six isolated role dashboards secured by JWT authentication and role-based access control.
- Support ticketing with real-time in-dashboard notifications.
- A Progressive Web App with offline caching and dynamic Malayalam translation.
- Serverless deployment on Vercel and Supabase for global availability and low operating cost.

Table 2.1 compares the existing and proposed systems.

Table 2.1: Comparison of Existing and Proposed Systems

Aspect	Existing System	Proposed System (TIMORA)
Discovery	Brokers and personal contacts	Public searchable marketplace
Pricing	Opaque, broker-controlled	Seller-set CFT price, visible to all
Measurement	Arbitrary quarter-girth use	Standard Hoppus and rectangular CFT
Compliance	Checked at checkposts	Form IV badge shown in advance
Trust	Informal reputation	Admin verification, ratings and reviews
Disputes	No formal channel	Ticketing with reference numbers
Access	Phone and physical visits	Responsive PWA, English and Malayalam

2.3 Use Case Diagram

The use case diagram in Figure 2.1 shows the actors of TIMORA and the services each can access. The Buyer, Seller / Mill Owner, Carpenter, Transport Owner and Super Admin are the primary actors. Operations that modify data first require authentication through the Login use case.

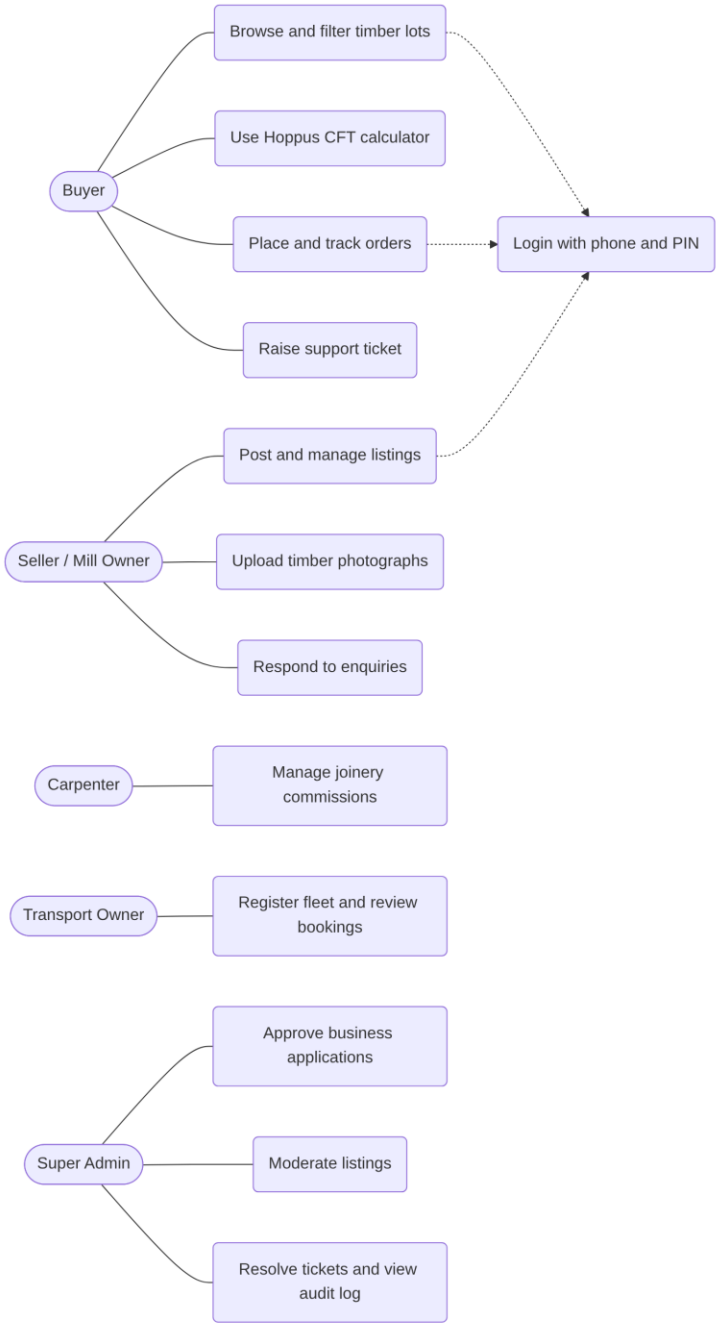


Figure 2.1: Use Case Diagram of TIMORA

2.4 Sequential Diagram

Figure 2.2 depicts the end-to-end interaction for the central business flow: a seller authenticates and publishes a lot, an administrator approves it, a buyer discovers it and places an order, and delivery is confirmed.

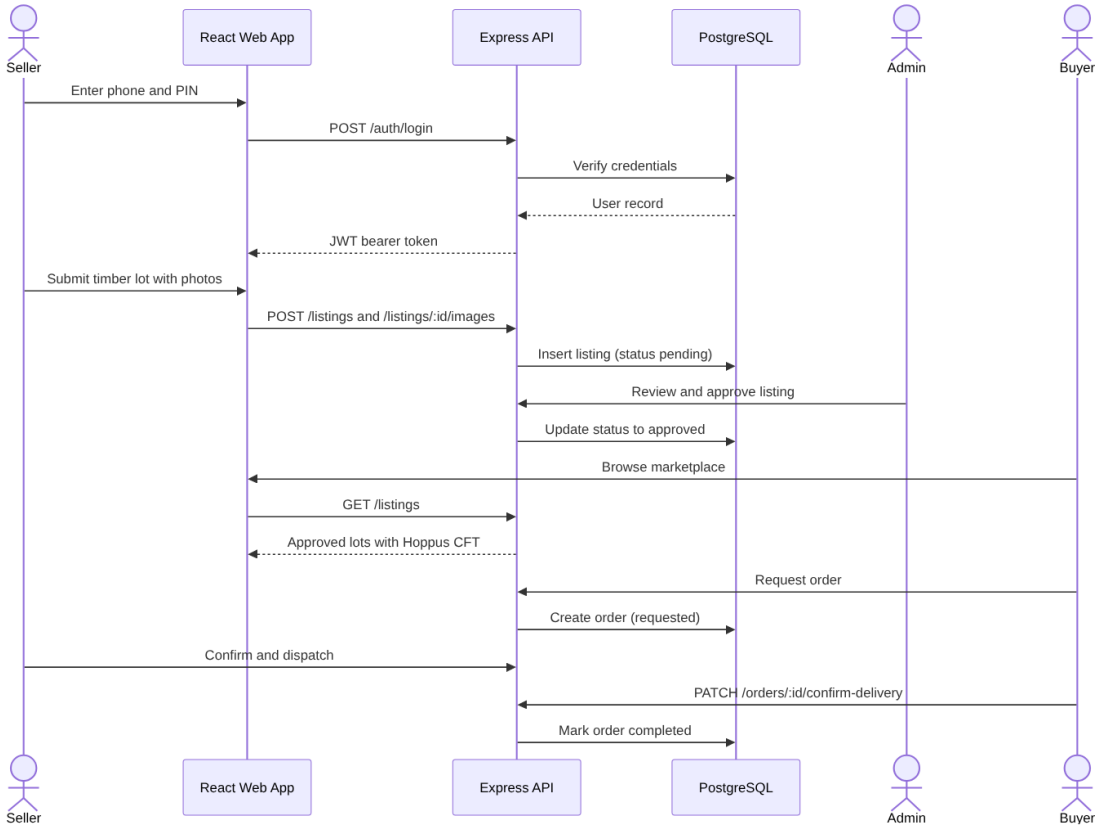


Figure 2.2: Sequence Diagram: Listing Publication to Delivery Confirmation

2.5 Activity Diagram

Figure 2.3 shows the activity flow of the listing lifecycle, including the automated spam check that rejects non-timber submissions and the moderation decision made by the administrator.

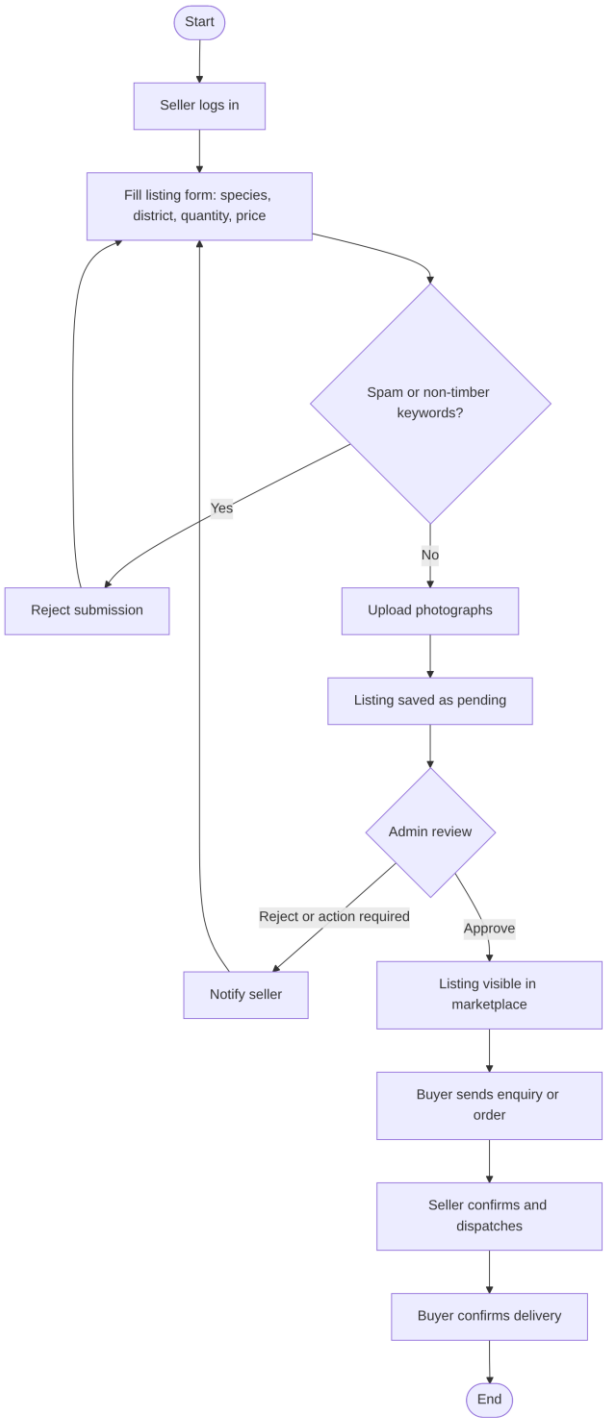


Figure 2.3: Activity Diagram: Timber Listing Lifecycle

CHAPTER NO 3 SYSTEM CONFIGURATION

3.1 Hardware Configuration

The TIMORA – Kerala's Smart Timber Marketplace project requires suitable hardware components to support application development, execution, testing and deployment. The hardware configuration includes the development machine used by the developers and the devices used by end-users to access the application. The following hardware components are considered for the project.

Processor: An Apple M-series processor, Intel Core i5 or an equivalent processor with at least four cores is recommended for development. It provides sufficient processing capability to run the code editor, development server, backend services and testing tools simultaneously.

Memory (RAM): A minimum of 8 GB RAM is required to support the development environment and execute the application. However, 16 GB RAM is recommended for smoother performance when running the front-end application, backend API, database-related tools and automated tests together.

Storage: A 256 GB Solid State Drive (SSD) is recommended for storing the project source code, development tools, dependencies and other related files. At least 5 GB of free storage space is required for maintaining the project monorepo and installing the necessary dependencies.

Display: A display resolution of 1366 × 768 pixels or higher is recommended for development activities. It provides sufficient screen space for viewing source code, application interfaces, debugging information and development tools.

Network Connection: A broadband internet connection is required during development for installing software packages, accessing online services, managing database connections and deploying the application. A stable network connection helps ensure smooth communication between the front end, backend API and Supabase database.

Client Device: TIMORA is designed to be accessed through smartphones, tablets and personal computers. A device with a minimum of 2 GB RAM and a current web browser is sufficient to access the application. The platform is also designed to support low-end Android devices, making it accessible to a wider range of users.

Web Hosting: The web application is hosted using the Vercel global edge Content Delivery Network (CDN). It delivers the static application files and Progressive Web Application (PWA) resources to users, helping improve accessibility and loading performance.

API Hosting: The backend API is deployed using Vercel serverless Node.js functions. These functions handle server-side operations such as authentication, timber listing management, order processing, support tickets and communication with the database.

Database and Storage: Supabase managed PostgreSQL is used for storing and managing application data. It maintains information such as user accounts, profiles, timber listings, orders and notifications. Supabase object storage is also used to store images uploaded for timber listings.

Internet Connectivity for End Users: Users can access TIMORA through 3G, 4G or Wi-Fi internet connections. The Progressive Web Application (PWA) caching mechanism helps retain essential application resources and supports improved accessibility during intermittent network connectivity.

These hardware components and hosting resources provide the necessary environment for developing, deploying and accessing the TIMORA application. The system is designed to operate across different devices while maintaining accessibility, responsiveness and reliable performance for users involved in Kerala's timber marketplace.

3.2 Software Configuration

The TIMORA – Kerala's Smart Timber Marketplace project is developed using various software tools and technologies that support application development, user interface design, backend processing, database management, testing and deployment. The following software components are used in the development process.

Operating System: The project can be developed and executed on macOS, Windows or Linux operating systems. These platforms provide the necessary environment for installing development tools, managing project files and running the application.

Node.js: Node.js version 18 or later is used as the runtime environment for executing the backend application. It supports server-side processing and enables the development and execution of the REST API.

npm and Turborepo: npm is used as the package manager for installing and managing project dependencies. Turborepo is used to organise and manage the monorepo structure, allowing the front end, backend, mobile application and shared packages to be maintained within a single project.

TypeScript: TypeScript version 5.7 is used as the primary programming language for application development. It provides static type checking and helps improve code consistency, maintainability and reliability across the different modules of the project.

React.js, Vite, Tailwind CSS and React Router: React 18 is used to develop the front-end user interface of TIMORA through reusable components. Vite 6 provides the development server and build environment, while Tailwind CSS 3.4 is used for designing responsive layouts and styling application components. React Router 6 manages navigation between different pages and supports role-based access to dashboards and other application modules.

Express.js: Express 4.21 is used as the backend framework for developing the REST API. It handles application routing, server-side operations, middleware and communication between the front end and database. It supports functionalities such as user authentication, timber listing management, order processing and administrative operations.

PostgreSQL and Supabase: PostgreSQL is used as the relational database management system, while Supabase provides the managed database environment and object storage facilities. These technologies are used to store and manage user details, profiles, timber listings, orders, support tickets and notification records. Supabase storage is also used to manage images uploaded to the marketplace.

React Native and Expo: React Native version 0.76 with Expo 52 is used to develop the mobile application shell for TIMORA. These technologies support cross-platform mobile application development for Android and iOS devices using TypeScript.

Vitest: Vitest is used as the testing framework to verify the functionality and reliability of the application. It supports unit and integration testing of calculation formulas, input validation, backend API operations and front-end components. It helps identify errors and ensure that important application features work as expected.

Git and GitHub: Git is used for version control and tracking changes made to the source code during development. GitHub is used to maintain the project repository and manage the code, making it easier to organise development activities and maintain different versions of the application.

Vercel and Supabase Deployment: Vercel is used to deploy and host the TIMORA web application and backend API. It provides global content delivery for the web application and supports serverless Node.js

functions for backend operations. Supabase is used to host the PostgreSQL database and provide image storage facilities.

Web Browser: Current versions of Google Chrome, Microsoft Edge, Mozilla Firefox and Safari are used to access and test the application. These browsers help verify the appearance, navigation, responsiveness and functionality of TIMORA across desktop computers, tablets and smartphones.

Visual Studio Code: Visual Studio Code is the code editor used for writing, editing and managing the project source code. It provides features such as syntax highlighting, extensions, debugging support and integrated terminal facilities, making it suitable for working with the different technologies used in TIMORA.

These software tools and technologies work together to provide a complete development and deployment environment for TIMORA. They support the implementation of a responsive web application, backend services, database operations, mobile application functionality, testing and secure deployment of the timber marketplace.

CHAPTER NO 4 SOFTWARE DESCRIPTION

The project is a monorepo of four workspaces. Table 4.1 lists each workspace with its technology and responsibility.

Table 4.1: Monorepo Workspaces and Technology Stack

Workspace	Technology Stack	Role and Responsibility
apps/web	React 18, Vite 6, Tailwind CSS 3.4, React Router 6, PWA	Responsive single-page web application with offline caching, Malayalam translation and loading skeletons
apps/api	Node.js 18+, Express 4.21, TypeScript, PostgreSQL (Supabase)	Headless REST API for JWT authentication, listings, support tickets and admin moderation
apps/mobile	React Native 0.76, Expo 52, TypeScript	Cross-platform mobile shell for Android and iOS
packages/shared	TypeScript 5.7, Vitest, Zod 3.23	Isomorphic library with CFT formulas, Hoppus calculators, currency formatters and Zod schemas

4.1 Overview Front End

The front end (apps/web) is a single-page application built with React 18 and bundled with Vite 6. React was chosen for its component model and large ecosystem, and Vite for its fast development server and optimised production builds.

- **React 18** renders the user interface from reusable components, using hooks for state, authentication context and session timers.
- **Vite 6** provides instant hot-module replacement during development and tree-shaken builds for production.
- **Tailwind CSS 3.4** supplies utility classes and a custom nature-inspired colour palette (see Section 5.3).

- **React Router 6** handles client-side routing, including role-based route guards for the dashboards and admin area.
- **Progressive Web App support** adds a service worker, offline caching and installability on mobile devices.
- **Dynamic Malayalam translation** exposes interface text and notifications in both English and Malayalam.
- **Loading skeletons and spinners** (Loader2 icon) give immediate feedback while network requests are in flight.
- **Session management:** the bearer token is retained across reloads, and a 25-minute inactivity timer opens a PIN re-authentication modal.

4.2 Overview Back End

The back end (apps/api) is a headless REST microservice written in TypeScript on Node.js 18+ and Express 4.21, deployed as serverless functions on Vercel. It persists data in a PostgreSQL database hosted on Supabase and stores listing photographs in a Supabase storage bucket. All endpoints are versioned under /api/v1.

The backend of the TIMORA – Kerala's Smart Timber Marketplace is developed using various technologies and security components to manage server-side operations, database communication, user authentication, input validation and data protection. The following components are used in the backend development process.

Express.js 4.21: Express.js is the backend framework used to develop the REST API of TIMORA. It handles HTTP routing, middleware processing and controllers. It manages requests received from the front end and performs the required server-side operations, including user authentication, timber listing management, order processing and administrative functions.

PostgreSQL (Supabase): PostgreSQL is used as the relational database management system, hosted through Supabase. It stores and manages application data while maintaining referential integrity between related tables. Row-level security is used to control data access, and atomic transactions help ensure the consistency of database operations.

JWT Bearer Tokens: JSON Web Tokens (JWT) are used to provide stateless user session authentication. After successful login, a token is generated and used to authenticate subsequent API requests. The tokens have a validity period of seven days, allowing the system to verify authenticated users and manage access to protected resources.

Argon2id: Argon2id is used for securely hashing users' six-digit PINs before storing them in the database. It applies salted password hashing to protect authentication credentials and prevents the original PIN from being directly stored or exposed.

Zod 3.23: Zod is used for validating incoming API requests and user input. It provides schema-based validation, with schemas shared between the front end and backend. This helps maintain consistent validation rules and ensures that the data submitted to the server follows the required format.

Helmet: Helmet is used to improve backend security by configuring appropriate HTTP security headers. These headers help protect the application against common web security threats, including cross-site scripting (XSS).

Express Rate Limit: Express Rate Limit is used to control the number of requests made to authentication endpoints. In TIMORA, it limits authentication requests to 100 requests within a 15-minute period. This helps reduce excessive requests and provides additional protection against repeated login attempts.

Parameterised Queries: Parameterised SQL queries are used when interacting with the PostgreSQL database. They separate SQL commands from user-supplied input, helping prevent SQL injection attacks and improving the security of database operations.

Vitest: Vitest is used for unit and integration testing of the backend components. It helps verify the functionality of API operations, authentication, input validation and database-related processes. It also helps identify errors and ensure that the backend services work as expected.

The shared package (packages/shared) is used by every workspace. Because the CFT and Hoppus formulas, currency formatters and validation schemas live in one place, a calculation performed in the browser, on the mobile app and on the server always yields the same figure.

CHAPTER NO 5 SYSTEM DESIGN

5.1 Architectural

TIMORA follows a three-tier, API-first architecture, shown in Figure 5.1. Clients (the web PWA and the Expo mobile shell) communicate only with the REST API. At the edge, the web application's vercel.json rewrites every /api/* request to the backend deployment. This avoids cross-origin (CORS) complications and mobile Safari cookie blocking, and lets the web app work without per-environment configuration.

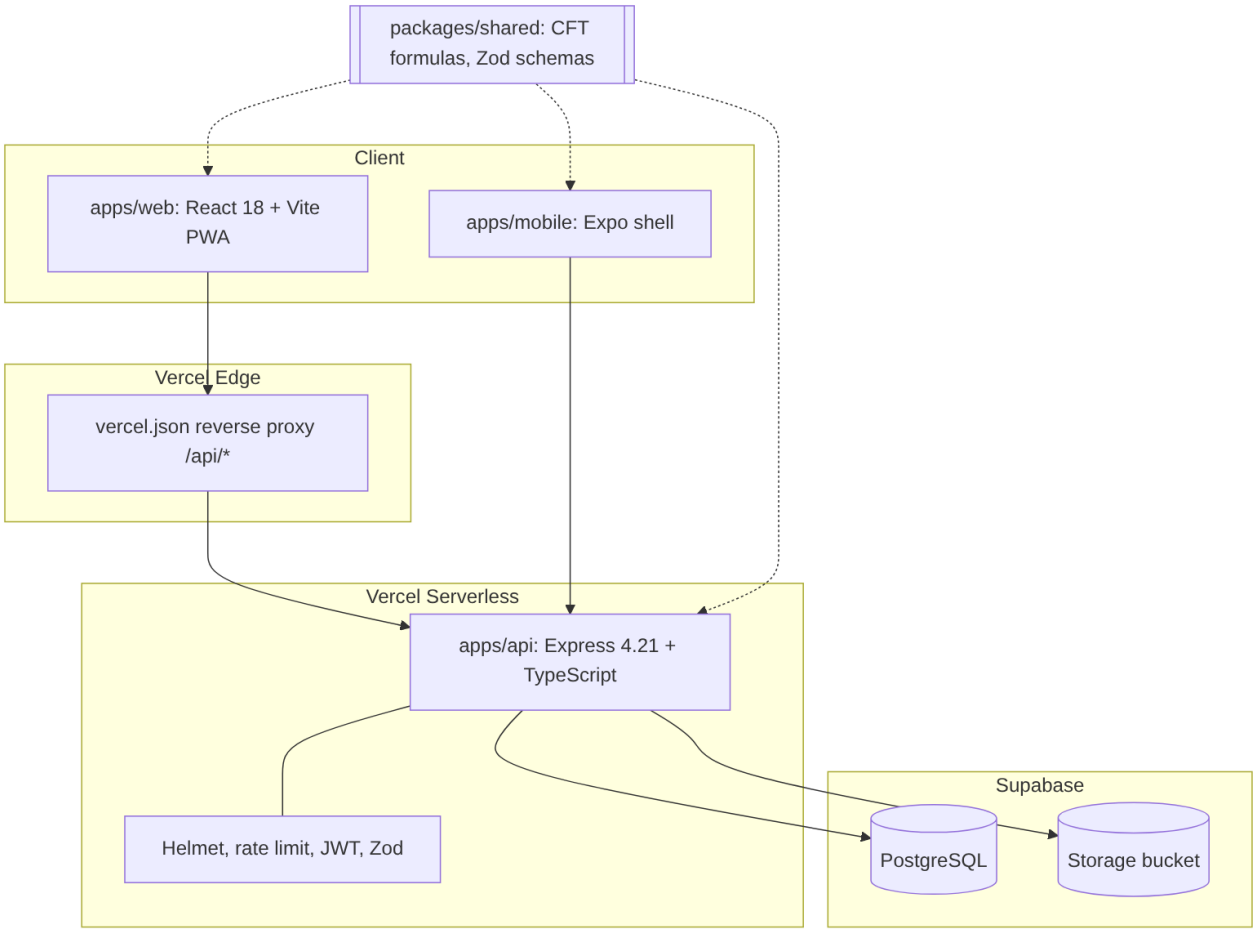


Figure 5.1: System Architecture of TIMORA

5.2 Input Design

Input forms are designed to collect only the data needed, validate it on the client and again on the server using shared Zod schemas, and give clear inline error messages. Phone numbers are limited to ten digits, PINs to six digits, and dimensions must be positive numbers. Table 5.2 summarises the main input screens.

Table 5.1: Principal Input Screens and Fields

Screen	Input Fields	Validation
Login	Mobile number, 6-digit PIN	Ten-digit phone; six-digit PIN; rate limited
Registration	Name, phone, PIN, role, district, business name, sawmill licence, GST number	Required fields by role; unique phone
Create Listing	Species, timber type, district, quantity, price per CFT, pricing mode, transit pass	Positive numbers; spam / non-timber keyword rejection
Image Upload	Multi-angle photographs, primary image flag	Image type and size limits
CFT Calculator	Girth, length, width, thickness, pieces, rate	Negative or zero dimensions rejected
Order Request	Quantity, agreed price, delivery address	Quantity within available stock
Support Ticket	Category, description, priority	Mandatory category and description
Admin Review	Decision, approval notes	Admin role only

5.3 Output Design

Outputs are presented as responsive cards, tables and dashboards in a minimal, nature-inspired visual language reflecting Kerala's rainforests, seasoned heartwood and sawdust yards. Table 5.3 lists the main outputs, and Table 5.4 defines the colour system used across the interface.

Table 5.2: Principal Output Screens

Output	Content
Marketplace listing grid	Lot cards with species, district, quantity, CFT price and Form IV badge
Lot detail page	Photo gallery, Hoppus CFT, seller details and enquiry action
Calculator result	Volume in CFT and cubic metres, estimated cost
Directory pages	Mills, carpenters and transporters with rates, capacity and reviews
Role dashboards	Orders, inventory, enquiries, commissions and fleet, by role
Notification feed	Bilingual alerts with action links and read status
Admin control cell	Pending applications, audit log and ticket queue

Table 5.3: Colour Palette and Design Tokens

Colour Token	Hex Code	Interface Role
Sand Beige	#E7E1D6	Primary background, surface containers, tag pills
Moss Green	#5B6147	Primary accent, buttons, active tabs, brand highlights
Deep Olive	#3F452F	Button hover, high-contrast headings, callouts
Dark Wood	#4A3626	Primary typography, logo text, species titles
Natural Wood	#7A5A3A	Secondary metadata, dimensional labels, captions
Warm Taupe	#8E7F6B	Card borders, input outlines, dividers

5.4 Database Design

TIMORA uses a relational PostgreSQL schema on Supabase that emphasises referential integrity, row-level security and atomic transactions. The entity-relationship diagram is shown in Figure 5.2, and the data dictionary of each table follows.

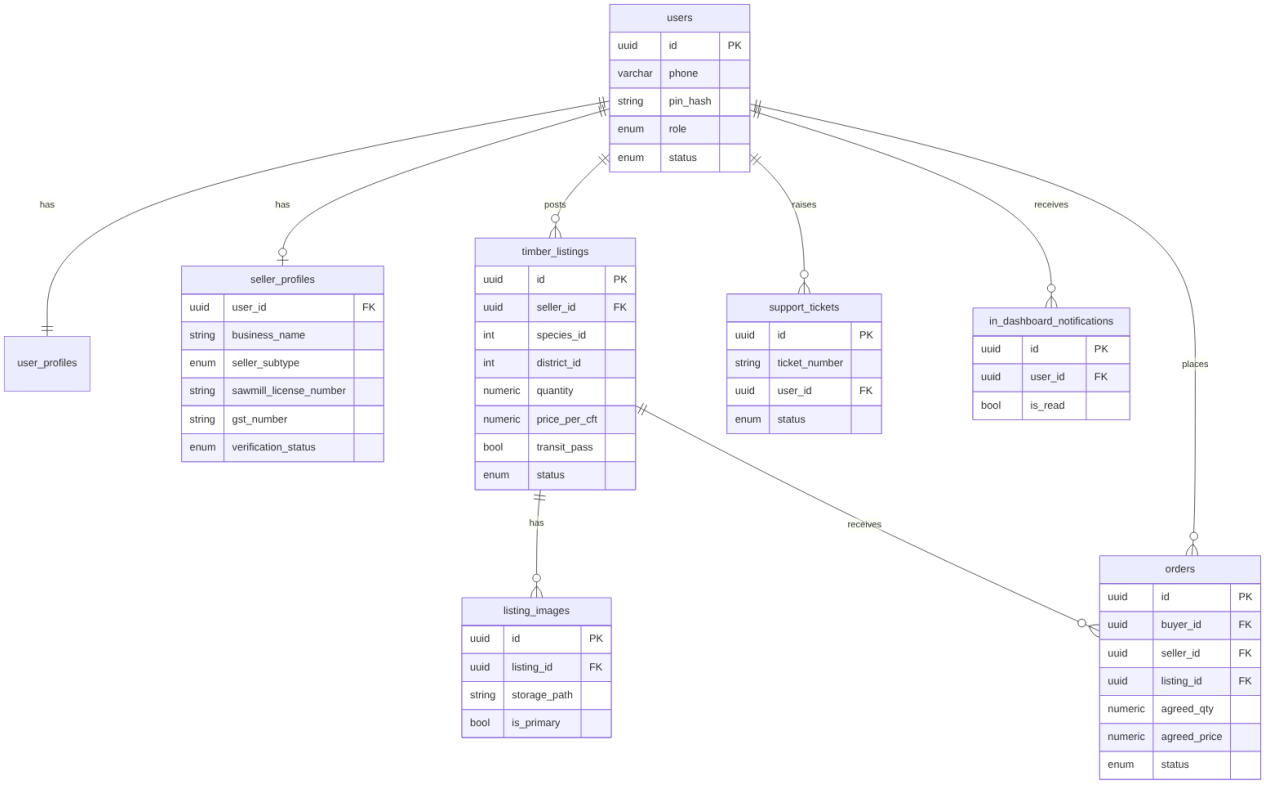


Figure 5.2: Entity-Relationship Diagram of TIMORA

Table 5.4: Table Structure: users

Column	Type	Description
id	UUID (PK)	Unique user identifier
phone	VARCHAR(10)	Registered mobile number (login ID)
pin_hash	TEXT	Argon2id hash of the PIN
role	ENUM	buyer, seller, carpenter, transport, admin
status	ENUM	active, suspended
created_at	TIMESTAMP	Account creation time

Table 5.5: Table Structure: user_profiles and seller_profiles

Table	Columns
user_profiles	user_id (FK), full_name, district, locality, email, notification preferences
seller_profiles	user_id (FK), business_name, seller_subtype (individual, mill_owner), sawmill_license_number, gst_number, verification_status, approval_notes

Table 5.6: Table Structure: timber_listings and listing_images

Table	Columns
timber_listings	id (PK), seller_id (FK), species_id, timber_type_id, district_id, quantity, price_per_cft, pricing_mode (fixed, negotiable), transit_pass (BOOLEAN), status (pending, approved, sold)
listing_images	id (PK), listing_id (FK), storage_path, public_url, is_primary (BOOLEAN), upload_timestamp

Table 5.7: Table Structure: orders

Column	Type	Description
id	UUID (PK)	Order identifier
buyer_id / seller_id	UUID (FK)	Parties to the consignment
listing_id	UUID (FK)	Timber lot ordered
agreed_qty / agreed_price	NUMERIC	Negotiated quantity and price
delivery_address	TEXT	Delivery destination
status	ENUM	requested, confirmed, dispatched, delivered_pending, completed
snapshot	JSON	Snapshot of the listing at order time

Table 5.8: Table Structure: support_tickets and in_dashboard_notifications

Table	Columns
support_tickets	ticket_number (e.g. TKT-2026-0042), user_id (FK), role, category, description, priority (normal, urgent), status (open, in_progress, resolved), resolution_reply
in_dashboard_notifications	user_id (FK), category, title_en, title_ml, body_en, body_ml, action_url, is_read (BOOLEAN)

CHAPTER NO 6 SOURCE CODE

This chapter presents representative excerpts of the core logic of TIMORA. The full source code is maintained in the project repository at https://github.com/rohancherian2003/Timora. The excerpts follow the formulas, limits and endpoints specified in the project documentation.

6.1 Shared Library: Hoppus and CFT Calculations

These functions live in packages/shared and are used by the web client, mobile app and API.

packages/shared/src/cft.ts

```
export const CFT_PER_CUBIC_METRE = 35.3147;
export const CUBIC_METRE_PER_CFT = 0.0283168;

/** Kerala Quarter-Girth Hoppus rule for round logs. */
export function hoppusCft(midGirthInches: number, lengthFeet: number): number {
  if (midGirthInches <= 0 || lengthFeet <= 0) {
    throw new Error('Dimensions must be positive');
  }
  const quarterGirth = midGirthInches / 4;
  return (quarterGirth * quarterGirth * lengthFeet) / 144;
}

/** Rectangular CFT for sawn planks, beams and battens. */
export function rectangularCft(
  lengthFeet: number, widthInches: number, thicknessInches: number, pieces = 1
): number {
  if ([lengthFeet, widthInches, thicknessInches, pieces].some((v) => v <= 0)) {
    throw new Error('Dimensions must be positive');
  }
  return (lengthFeet * widthInches * thicknessInches * pieces) / 144;
}

export const cftToCubicMetres = (cft: number) => cft * CUBIC_METRE_PER_CFT;
```

6.2 Shared Library: Validation Schemas

packages/shared/src/schemas.ts

```
import { z } from 'zod';

export const phoneSchema = z.string().regex(/^\d{10}$/, "Enter a 10-digit mobile number");
export const pinSchema = z.string().regex(/^\d{6}$/, "PIN must be 6 digits");

export const loginSchema = z.object({ phone: phoneSchema, pin: pinSchema });

export const listingSchema = z.object({
  speciesId: z.number().int().positive(),
  districtId: z.number().int().positive(),
  quantity: z.number().positive(),
  pricePerCft: z.number().positive(),
  pricingMode: z.enum(['fixed', 'negotiable']),
  transitPass: z.boolean(),
});
```

6.3 API: Application Bootstrap and Security Middleware

apps/api/src/app.ts

```
import express from 'express';
import helmet from 'helmet';
import rateLimit from 'express-rate-limit';
import { authRouter } from './routes/auth';
import { listingsRouter } from './routes/listings';

const app = express();
app.use(helmet());
app.use(express.json());

const authLimiter = rateLimit({ windowMs: 15 * 60 * 1000, max: 100 });
app.use('/api/v1/auth', authLimiter, authRouter);
app.use('/api/v1/listings', listingsRouter);

export default app;
```

6.4 API: Login Controller

apps/api/src/controllers/auth.ts (excerpt)

```
import argon2 from 'argon2';
import jwt from 'jsonwebtoken';
import { loginSchema } from '@timora/shared';

export async function login(req, res) {
  const { phone, pin } = loginSchema.parse(req.body);
  const { rows } = await db.query('SELECT * FROM users WHERE phone = $1', [phone]);
  const user = rows[0];
  if (!user || !(await argon2.verify(user.pin_hash, pin))) {
    return res.status(401).json({ error: 'Invalid phone or PIN' });
  }
  const token = jwt.sign({ sub: user.id, role: user.role },
    process.env.JWT_SECRET!, { expiresIn: '7d' });
  res.json({ token, role: user.role });
}
```

6.5 Web: Edge Reverse Proxy Configuration

apps/web/vercel.json

```
{
  "rewrites": [
    { "source": "/api/:path*",
      "destination": "https://timora-backend-three.vercel.app/api/:path*" },
    { "source": "/(.*)", "destination": "/index.html" }
  ]
}
```

6.6 Web: Inactivity Timeout

apps/web/src/hooks/useInactivityTimeout.ts

```
import { useEffect } from 'react';

const TIMEOUT_MS = 25 * 60 * 1000; // 25 minutes

export function useInactivityTimeout(onTimeout: () => void) {
  useEffect(() => {
    let timer = setTimeout(onTimeout, TIMEOUT_MS);
    const reset = () => { clearTimeout(timer); timer = setTimeout(onTimeout, TIMEOUT_MS); };
    const events = ['click', 'keydown', 'mousemove', 'touchstart'];
    events.forEach((e) => window.addEventListener(e, reset));
    return () => { clearTimeout(timer); events.forEach((e) => window.removeEventListener(e, reset)); };
  }, [onTimeout]);
}
```

CHAPTER NO 7 SYSTEM TESTING

Testing was carried out continuously throughout development. The project maintains an automated suite of 186 Vitest tests across all workspaces, all of which pass. The suite covers unit tests of the shared formulas, integration tests of the API and component and route tests of the web application.

Table 7.1: Automated Test Suites

Test Suite	Files / Tests	Verification Scope
apps/api (Backend)	14 files, 138 tests	JWT authentication, PIN validation, rate limiting, RLS permissions, seller verification, race-condition prevention, order transactions
apps/web (Frontend)	6 files, 40 tests	Route guards, authentication retention, session timeout modal, marketplace filters, directory navigation, loading skeletons
packages/shared	1 file, 8 tests	Hoppus accuracy, rectangular CFT, negative-dimension rejection, phone validators
Total	21 files, 186 tests	100% passing

The test levels applied were unit testing (formulas and validators), integration testing (API with database), functional testing (role workflows) and security testing (authentication, rate limiting and access control).

7.1 Testcase

Table 7.2 lists the principal test cases with their expected and actual outcomes. Worked values for the calculator tests follow the formulas in Section 6.1: a 48 in girth and 12 ft log gives $(48/4)^2 \times 12 / 144 = 12$ CFT, and one 10 ft × 6 in × 2 in plank gives 120 / 144 = 0.83 CFT.

Table 7.2: System Test Cases

ID	Test Case	Input	Expected Result	Status
TC01	Valid login	Registered phone, correct 6-digit PIN	JWT issued; redirect to role dashboard	Pass
TC02	Wrong PIN	Valid phone, incorrect PIN	401 error; no token	Pass
TC03	Invalid PIN format	PIN "12ab"	Validation error	Pass
TC04	Auth rate limit	> 100 requests in 15 min	Requests throttled	Pass
TC05	Route guard	Buyer opens /admin	Access denied / redirect	Pass
TC06	Session timeout	25 minutes inactivity	PIN re-authentication modal shown	Pass
TC07	Hoppus calculation	Girth 48 in, length 12 ft	12.00 CFT	Pass
TC08	Rectangular CFT	10 ft × 6 in × 2 in × 1	0.83 CFT	Pass
TC09	Negative dimension	Length −5	Input rejected	Pass
TC10	Metric conversion	35.3147 CFT	1.00 m^3	Pass
TC11	Create listing	Valid seller form	Listing saved as pending	Pass
TC12	Spam detection	Title "Teak sofa set"	Submission rejected	Pass
TC13	Species filter	Select Teak pill	Only teak lots shown	Pass

ID	Test Case	Input	Expected Result	Status
TC14	Unapproved listing	Buyer browses marketplace	Pending lots not shown	Pass
TC15	Admin approval	Admin approves business	Seller status changes to APPROVED	Pass
TC16	Concurrent orders	Two buyers order the last lot	Only one order succeeds	Pass
TC17	Confirm delivery	Buyer PATCH confirm-delivery	Order status completed	Pass
TC18	Support ticket	Submit dispute	Reference TKT-YYYY-NNNN created	Pass
TC19	Ticket resolution	Admin resolves ticket	Notification appears in user feed	Pass
TC20	Button loading state	Click Publish	Button disabled with spinner	Pass

CHAPTER NO 8 SYSTEM IMPLEMENTATION / DEPLOYMENT

TIMORA was implemented in iterative phases: shared library and database schema, REST API, web client, role dashboards and moderation, testing and hardening, and finally production deployment. The production platform is live, as summarised in Table 8.1.

Table 8.1: Production Deployments

Service Tier	URL	Status
Web Frontend (SPA / PWA)	https://timora-rho.vercel.app/	Active (Vercel Global Edge CDN)
Backend REST API	https://timora-backend-three.vercel.app/	Active (serverless Node.js)
API Health and Meta	https://timora-backend-three.vercel.app/api/v1	Active (HTTP 200 OK)

Figure 8.1 shows the deployment pipeline. Pushing to the GitHub repository triggers Vercel builds for the web and API workspaces. The web build is distributed through the edge CDN, and its reverse proxy forwards /api/* calls to the serverless API, which connects to Supabase.

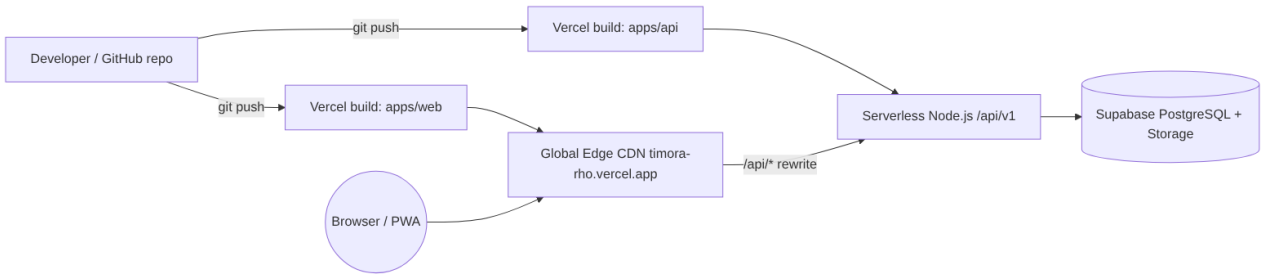


Figure 8.1: Deployment Pipeline of TIMORA

8.1 Local Development Setup

1. Clone the repository: git clone https://github.com/rohancherian2003/Timora.git and change into the folder.
2. Install dependencies with npm install; Turborepo links all apps and packages.
3. Configure environment variables (DATABASE_URL, JWT_SECRET, SUPABASE_URL, SUPABASE_KEY) in apps/api/.env.
4. Run migrations and seed data: npm run db:migrate --workspace=apps/api, then npm run db:seed --workspace=apps/api.
5. Start the development servers with npm run dev (web on port 5173, API on port 3001).
6. Run the 186-test suite with npm run test.

8.2 Production Deployment Notes

- Secrets such as JWT_SECRET, database credentials and the administrator PIN are provided as environment variables and are never committed to the repository.
- The production database is seeded with approved demo accounts for each role to support evaluation; their credentials are supplied separately and are not reproduced in this report.
- The development OTP flow is a demonstration mode; a production SMS provider will be integrated and approved separately before public launch.
- The system is designed in harmony with the Kerala Forest Produce Transit Rules, 1975 (movement of timber across district boundaries and checkposts), Form IV transit permits for reserved species such as Teak and Rosewood from private land, sawmill establishment permits (licence numbers recorded and verified by moderators before public quoting is allowed) and KFRI Hoppus conventions for taper and girth measurement.

CHAPTER NO 9 CONCLUSION AND FUTURE ENHANCEMENT

9.1 Conclusion

TIMORA demonstrates that Kerala's fragmented timber trade can be brought onto a single transparent digital platform. The system connects buyers, sellers, mills, carpenters and transporters, standardises volume measurement through the Kerala Hoppus rule, surfaces Form IV compliance and provides moderated, role-isolated workflows. Built on a modern TypeScript monorepo and deployed on Vercel and Supabase, it is live, secured by Argon2id, JWT, Helmet and rate limiting, and verified by 186 passing automated tests.

The project met its objectives of direct discovery, measurement standardisation, compliance visibility, role-based operations and dispute handling, and provides a practical foundation for a commercially operated timber marketplace.

9.2 Future Enhancement

- Online payment integration with escrow-style release on delivery confirmation.
- Real-time delivery and shipment tracking for transport operators.
- Form IV document upload and automated verification against forest department records.
- Production SMS / OTP provider integration to replace the demonstration OTP.
- Full development and store release of the Expo-based Android and iOS application.
- Price analytics, market-rate trends and recommendations for growers.
- Expansion beyond Kerala to neighbouring states and additional wood species.
- Additional regional-language support and voice-assisted listing creation.

CHAPTER NO 10 APPENDICES

10.1 Screenshots

The following figures present the main screens of the live application at https://timora-rho.vercel.app/. Each figure shows the corresponding screen.

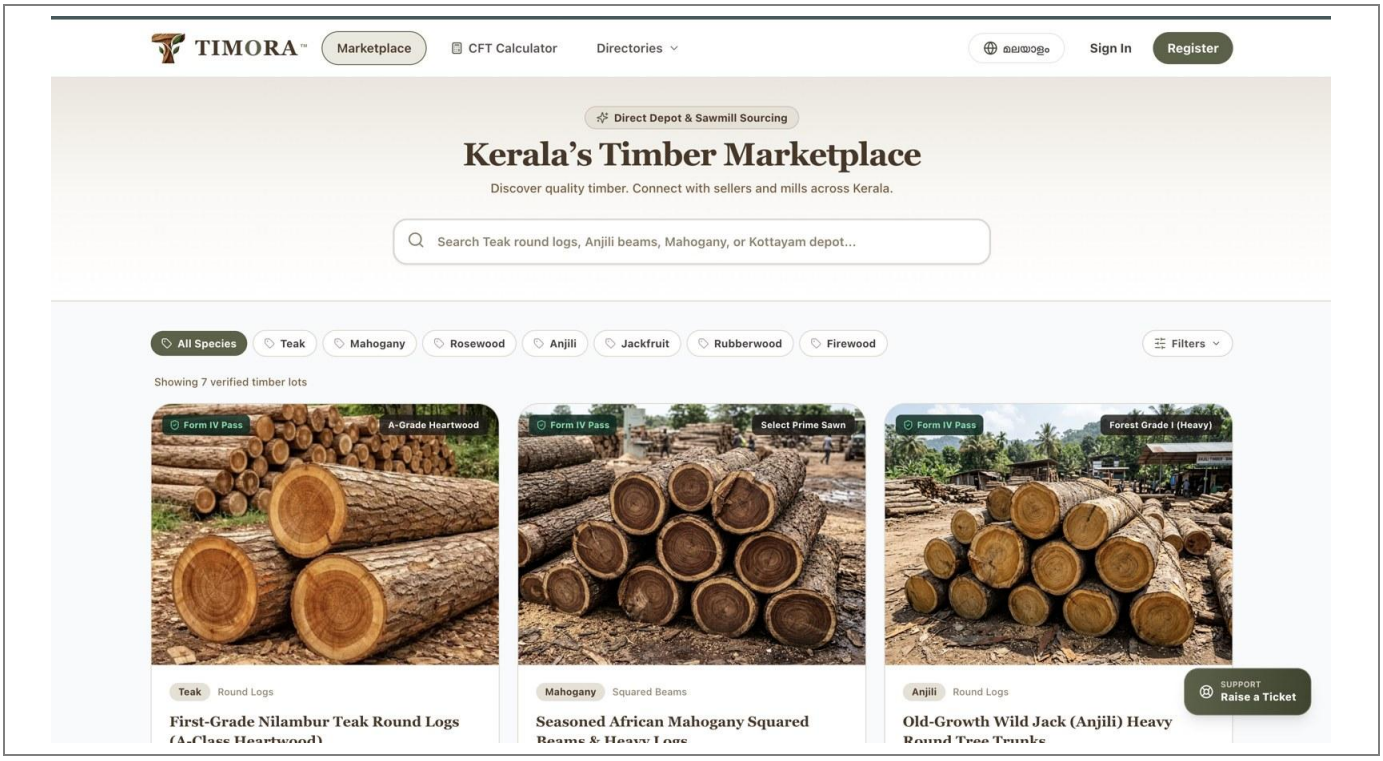


Figure 10.1: Home Page

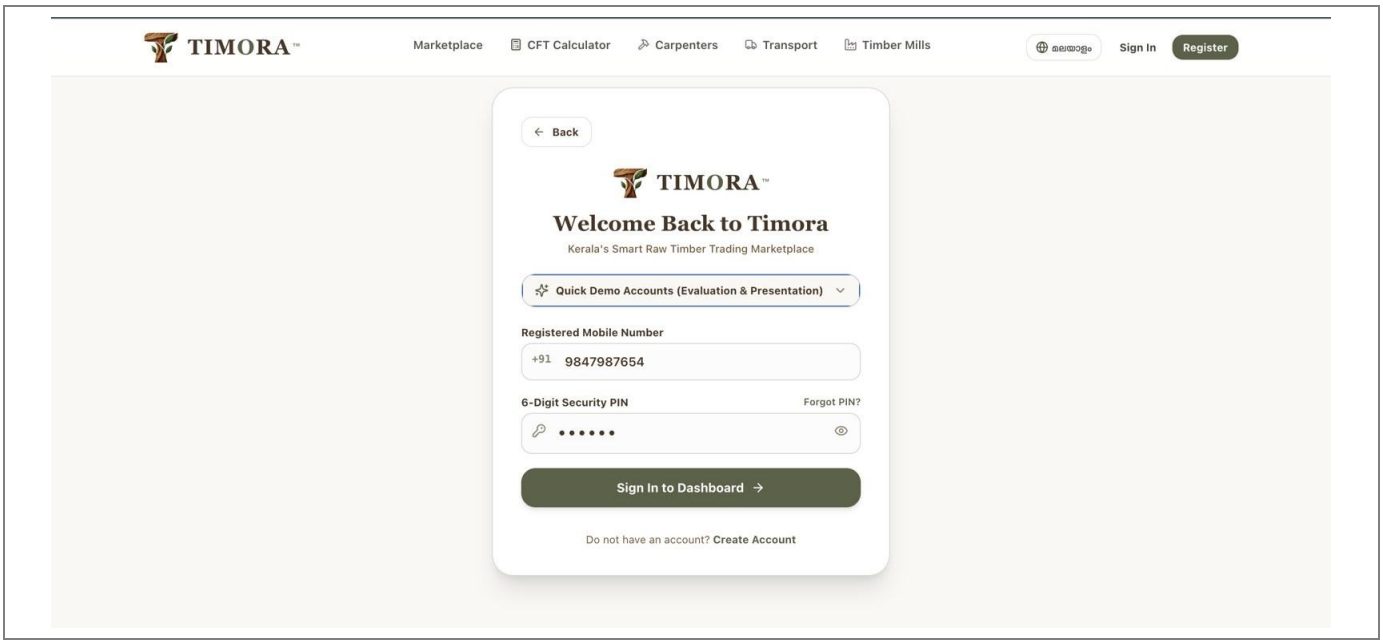


Figure 10.2: Login Screen with Quick-Demo Buttons

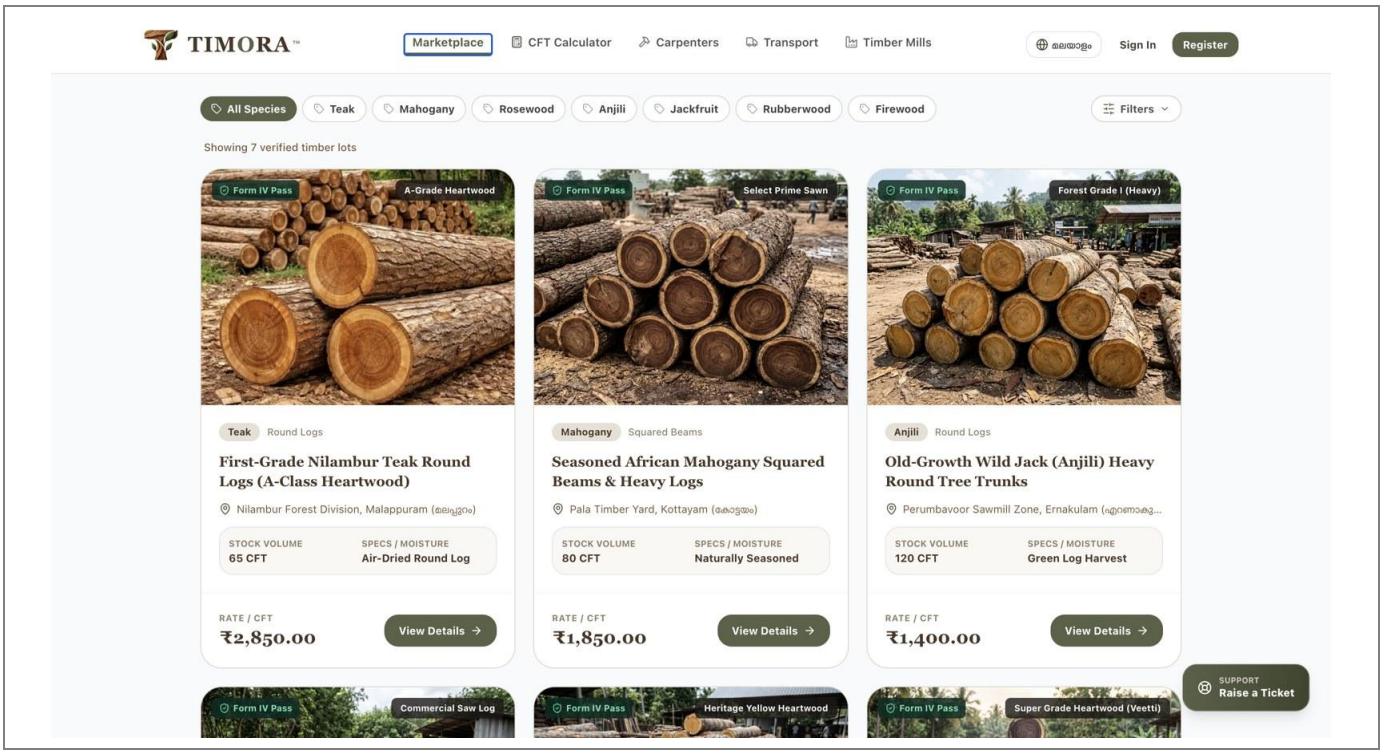


Figure 10.3: Raw Timber Stock Exchange (Marketplace)

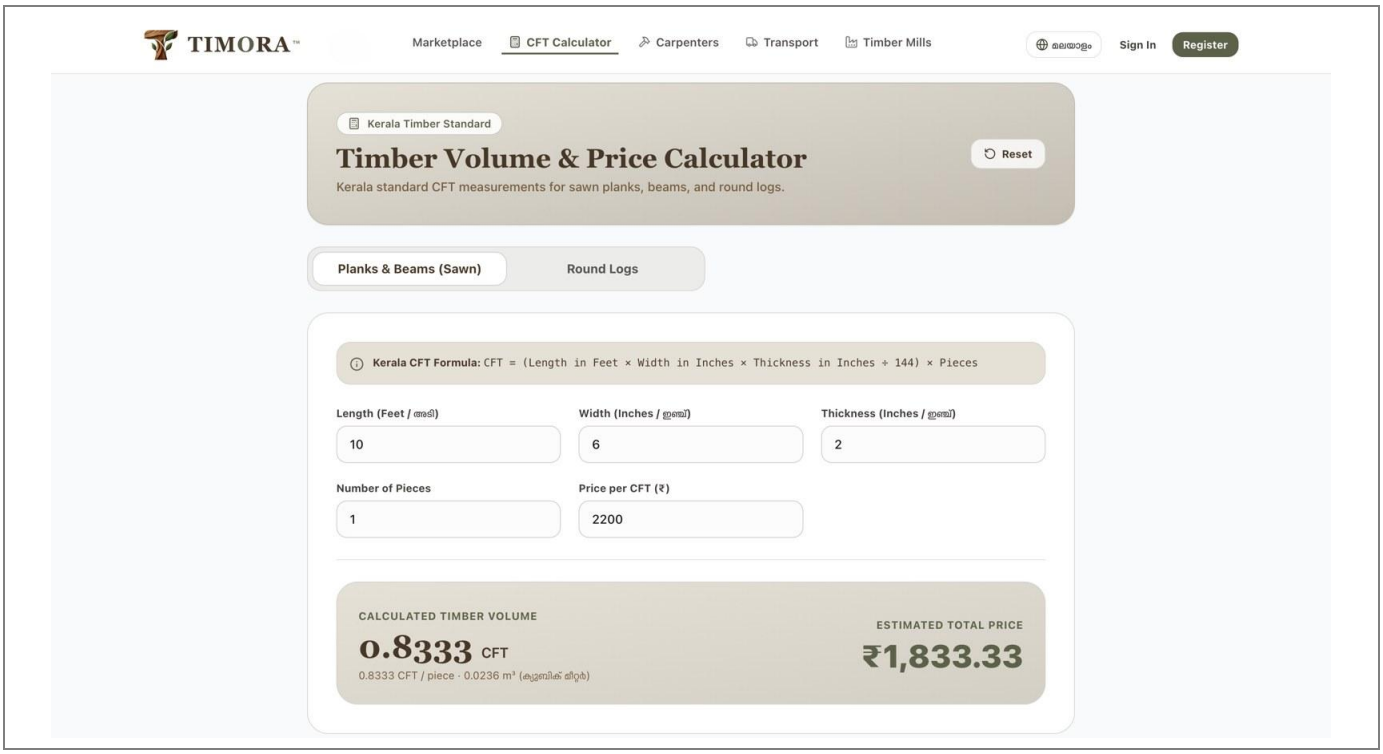


Figure 10.4: Hoppus CFT Calculator

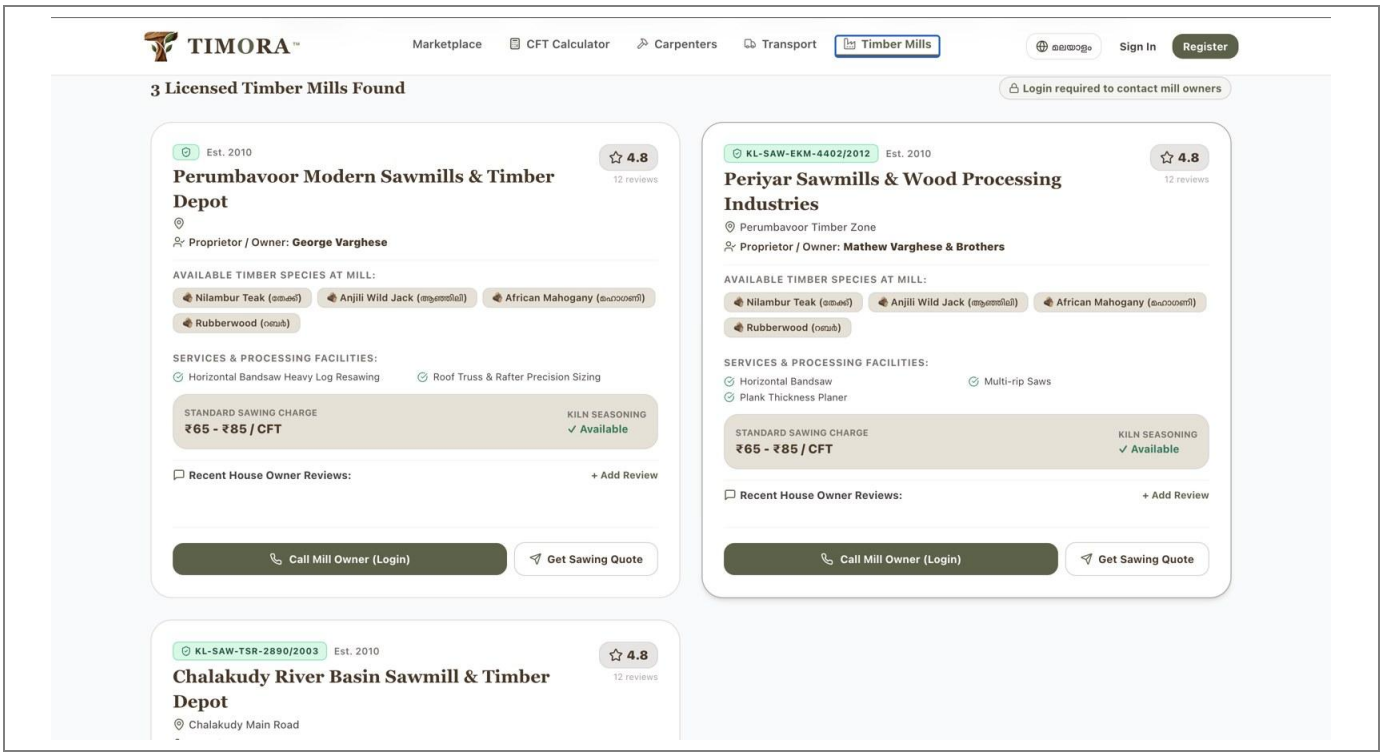


Figure 10.5: Licensed Sawmills Directory

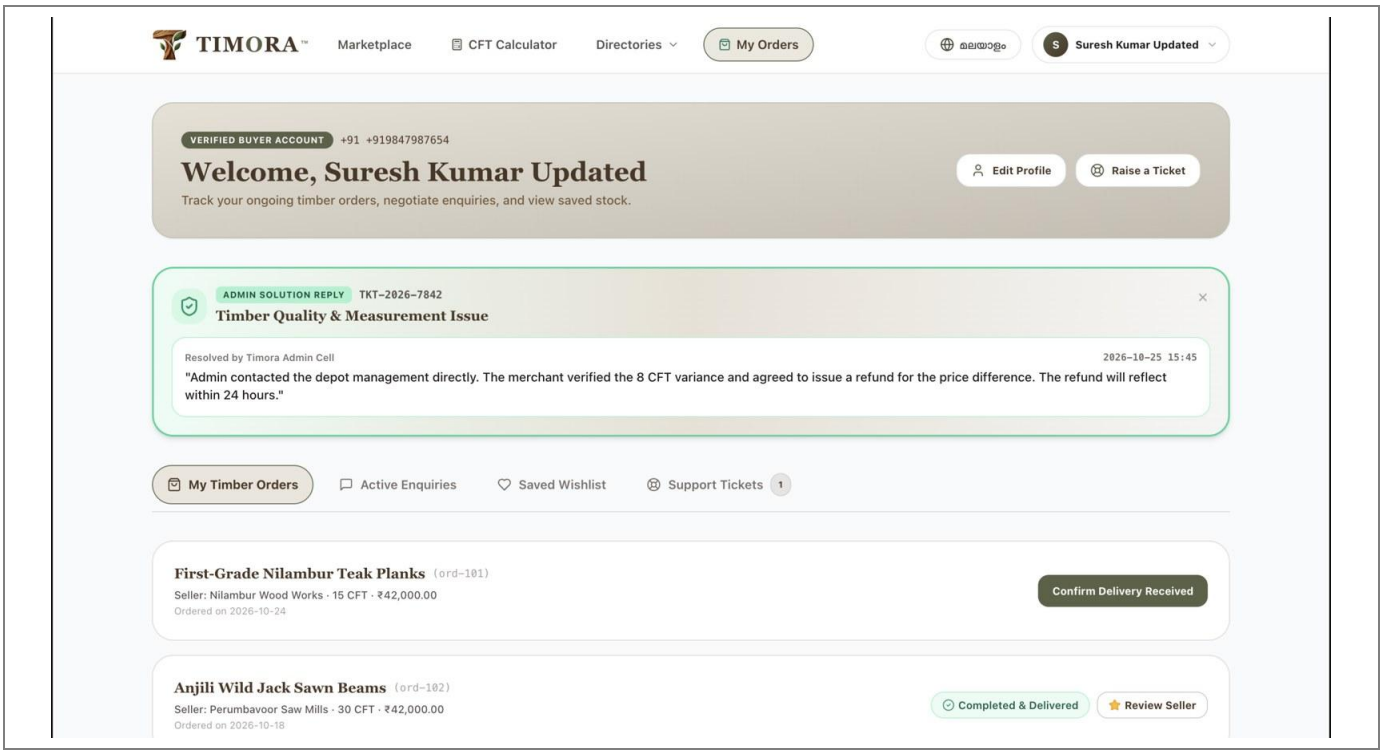


Figure 10.6: Buyer Dashboard

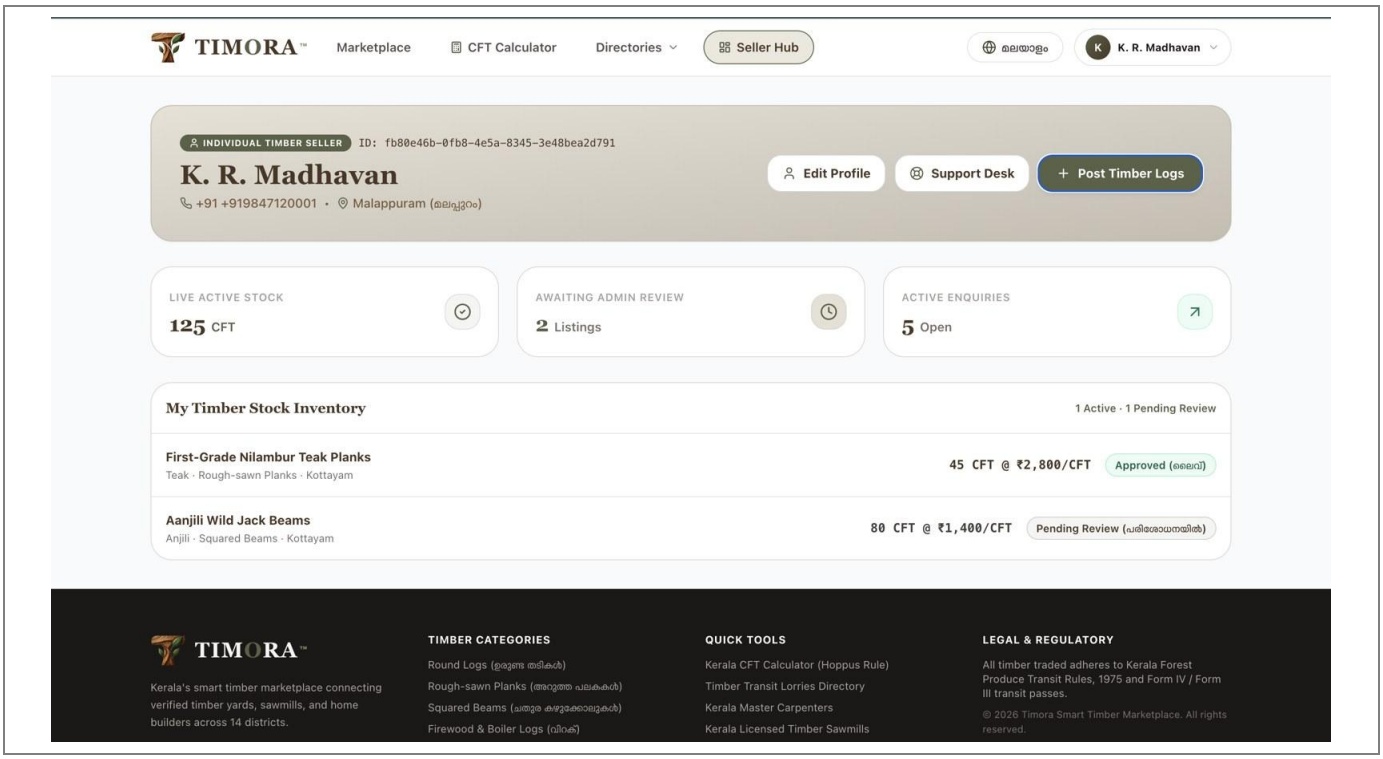


Figure 10.7: Seller Dashboard and Listing Form

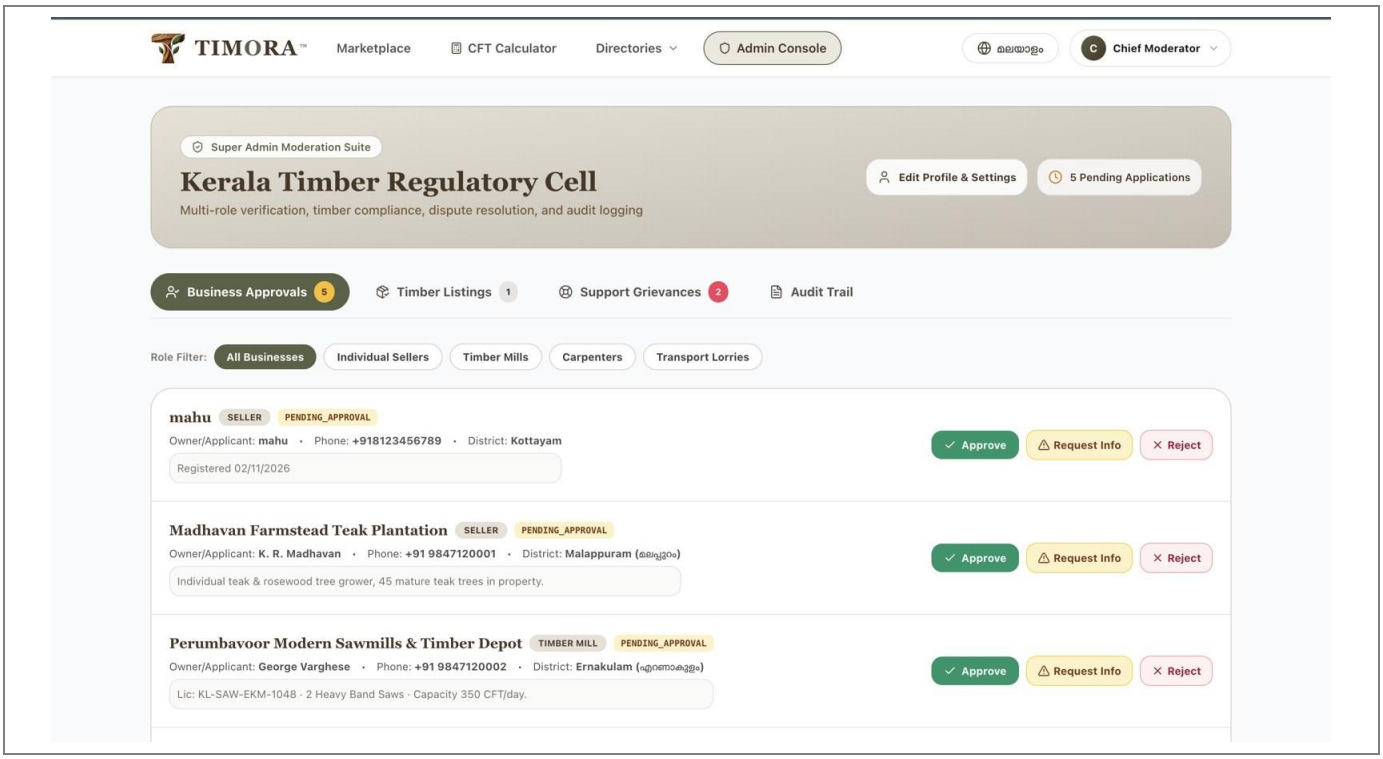


Figure 10.8: Super Admin Control Cell

10.2 Additional Reference Material

Table 10.1 summarises the volume formulas and conversion constants used by the platform.

Table 10.1: Volume Formulas and Constants

Item	Specification
Hoppus rule (round logs)	Volume (CFT) = [(Mid-girth in inches / 4)² × Length in feet] / 144
Rectangular CFT (sawn timber)	Volume (CFT) = [Length (ft) × Width (in) × Thickness (in) × Pieces] / 144
Cubic metre to CFT	1 m³ = 35.3147 CFT
CFT to cubic metre	1 CFT = 0.0283168 m³

Role to dashboard routes: Timber Buyer (/dashboard/buyer), Individual Seller (/dashboard/seller), Timber Mill Owner (/dashboard/mill-owner), Master Carpenter (/dashboard/carpenter), Transport Owner (/dashboard/transport) and Super Admin (/admin).

CHAPTER NO 11 REFERENCES / BIBLIOGRAPHY

[1] Meta Platforms, "React Documentation," https://react.dev.

[2] Vite, "Vite Guide," https://vite.dev/guide/.

[3] Tailwind Labs, "Tailwind CSS Documentation," https://tailwindcss.com/docs.

[4] React Router, "React Router Documentation," https://reactrouter.com.

[5] OpenJS Foundation, "Express.js Documentation," https://expressjs.com.

[6] Supabase, "Supabase Documentation (Postgres, Row Level Security, Storage)," https://supabase.com/docs.

[7] Vercel, "Vercel Documentation (Rewrites and Serverless Functions)," https://vercel.com/docs.

[8] Turborepo, "Turborepo Documentation," https://turbo.build/repo/docs.

[9] Expo, "Expo Documentation," https://docs.expo.dev.

[10] Zod, "Zod Documentation," https://zod.dev.

[11] Vitest, "Vitest Documentation," https://vitest.dev.

[12] A. Biryukov, D. Dinu and D. Khovratovich, "Argon2: the memory-hard function for password hashing," RFC 9106, IETF, 2021.

[13] M. Jones, J. Bradley and N. Sakimura, "JSON Web Token (JWT)," RFC 7519, IETF, 2015.

[14] OWASP Foundation, "OWASP Top Ten Web Application Security Risks," https://owasp.org/www-project-top-ten/.

[15] Government of Kerala, "Kerala Forest Produce Transit Rules, 1975."

[16] Kerala Forest Research Institute (KFRI), timber measurement and Hoppus volume conventions.

[17] R. K. Cherian, "TIMORA: Comprehensive Technical Architecture and System Documentation," Version 1.0.0, November 2026.